The World Model as Judge: Learned Traversal Rewards for Humanoid Navigation in Cluttered, Occupied Spaces

Target venues: ICRA 2027 (robotics framing) / ICLR 2027 (reward-learning framing).
Status: working draft; numbers marked `[v3]` / `[P2]` land from the running pipeline.

Abstract

Humanoid robots can walk, but they cannot yet traverse the narrow, cluttered, human-occupied spaces where they are meant to work. The obstacle is not low-level control — whole-body controllers trained in physics simulation are increasingly capable — but **reward specification**: behaviors such as turning shoulders through a tight gap, crouching under furniture, or yielding to a person resist hand-crafted reward engineering, and their quality is inherently perceptual. We propose using a **video-language world model as a learned reward model**. We fine-tune a compact (2B) video-language model into a traversal critic that scores short navigation clips on collision safety, clearance management, motion quality, and social compliance, supervised by privileged simulator state that is automatically converted to ordinal labels — no human annotation. The critic then provides reward shaping for training a lightweight vision-based navigation policy that commands a frozen, pretrained whole-body controller; the world model is used **at training time only**, leaving deployment unencumbered. On held-out scenes, the fine-tuned critic predicts ground-truth traversal quality from pixels alone (Pearson r = 0.48 vs. 0.02 before fine-tuning), improving monotonically with data scale.
`[P2: Critic-shaped policies achieve X% fewer collisions and Y-point higher traversal-quality scores at equal success rate versus hand-crafted-reward baselines.]`

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1. Introduction

1.1 The story in one paragraph

The last three years solved humanoid locomotion: massively parallel physics-simulation RL now produces whole-body controllers that walk, crawl, and recover from pushes, and behavior foundation models such as GEAR-SONIC compress thousands of hours of human motion into a single deployable policy. What it did not solve is where those robots are supposed to go: through the 0.6-meter gap between the sofa and the bookshelf, under the low table, past the person carrying groceries. This is not a control problem — the controller already knows how to crouch — it is a **decision and reward problem**. Nobody can write down the reward function for "traverse this room the way a considerate person would," and every hand-crafted proxy (clearance penalties, velocity bonuses, personal-space costs) is a brittle stand-in that RL exploits. We show that a video world model, fine-tuned cheaply on automatically labeled simulation clips, can be that reward function — and that unlike its hand-crafted competitors, it reads quality from pixels, so it generalizes to scenes, viewpoints, and (ultimately) real videos where no privileged state exists.

1.2 The observation the paper is built on

Generative video world models are used two ways in robotics today: as **simulators** (roll out imagined futures for planning) and as **policies** (action-conditioned VLAs). Both uses stress exactly what these models are worst at — long-horizon pixel-consistent generation — and both put a large model in or near the control loop. We exploit the third option: a video model as a **judge**. Recognition is easier than generation; scoring "was that traversal safe and natural?" needs one VLM forward pass over a short clip, not thirty diffusion steps per imagined frame. Judging is also the only use that is **training-time only**: the deployed system is a ~0.5M-parameter policy on top of a frozen controller, with no world model on the robot.

1.3 Contributions

1. **A method**: fine-tune a compact video-language world model into a traversal critic — an ordinal (1–5) scorer of navigation clips — supervised entirely by privileged simulator state via a deterministic auto-labeler (four interpretable axes: safety, clearance, social, progress; safety as hard ceiling). No human labels; supervision scales with simulation.
2. **Evidence the premise holds**: on scenes held out at the scene level (novel layouts, gap widths, furniture, appearance), the critic recovers ground-truth traversal quality from pixels alone: Pearson r = 0.02 (pretrained, zero-shot) → 0.38 (460 training clips) → **0.48** (1,530 clips), acc±1 0.755 — a clean monotone scaling curve. `[v3: balanced ~2,900-clip result]`
3. **A system**: a three-layer architecture (frozen SONIC whole-body controller at 50 Hz; small vision policy at ~3 Hz; critic as asynchronous training-time reward) with an async scoring protocol that never blocks RL on the 2B model. `[P2: paired-seed comparison — critic-shaped vs. hand-crafted-reward PPO on held-out scenes.]`
4. **A cautionary datapoint for the field**: our own hand-crafted baseline reward collapsed the policy to standing still on the first attempt (per-frame contact penalties dominated progress) — an accidental, quantified illustration of exactly the brittleness the critic is designed to remove.

1.4 What this paper is not

We do not claim the critic replaces task reward: sparse goal reward and privileged safety terms remain the backbone; the critic shapes. We do not claim video world models should replace simulators. And the current experiments are in simulation with kinematic playback of a pretrained motion planner; physics-in-the-loop training and real-robot deployment are staged as follow-up (§7).

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2. Related work (skeleton)

- **Humanoid whole-body control / behavior foundation models.** GEAR-SONIC, Decoupled WBC (GR00T), BeyondMimic, unitree_rl_lab. We consume these frozen.
- **Cluttered-space humanoid traversal.** HumanoidPF / Click-and-Traverse (G1, MuJoCo, procedural obstacles with difficulty knobs) — closest prior work; static clutter,

hand-crafted rewards. Our deltas: dynamic people, learned perceptual reward, style-mode action space.

- **Social navigation.** Classic (ORCA, social forces), learned (NavIsaacLab crowds, Habitat 3.0 social nav). Reward is invariably hand-crafted; our critic is the missing learned-reward piece.
- **World models in robotics.** As simulators (dreamer-style, video prediction for MPC), as policies (VLA: GR00T N-series, π0, Cosmos policy mode). As judges: VideoPhy-2-style physical-plausibility scoring is the nearest neighbor — we extend the idea from "is this video physical?" to "is this traversal good?" and close the loop into RL.
- **Learned rewards / RLHF / VLM-as-reward.** Reward models from preferences, VLM zero-shot rewards (e.g., CLIP-based), foundation-model feedback for RL. Our distinction: video (not image) judgment, ordinal calibrated output, auto-labels from privileged sim state rather than human preference.

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3. Method

3.1 Problem setup

Goal-conditioned navigation in cluttered indoor scenes with moving people. Deployed stack: frozen whole-body controller **C** (SONIC: latent-token motion tracking at 50 Hz + kinematic planner exposing {mode, direction, speed, height}); small vision policy **π** (egocentric RGB + goal vector → planner commands at ~3 Hz). Train π with PPO; the research question is the reward.

3.2 Traversal critic

Data. Parameterized procedural scenes (doorway gaps 0.6–1.2 m, furniture, low tables, moving people on crossing paths; per-scene visual randomization). Scripted policies spanning five quality bands generate diverse rollouts. Privileged channels recorded per frame: min clearance (analytic point-to-box), min person distance, base speed, goal distance, contact flags.

Auto-labeler. Four axis scores from documented thresholds — safety (contact duration), clearance (5th-percentile margin + slow-when-tight bonus), social (min person distance), progress (goal-rate + freeze penalty) — composed as overall = min(safety, half-down-round(mean)). Safety is a hard ceiling: a collision episode cannot score well on style. Scene-hash split: validation scenes are entirely unseen (layout and appearance).

Model. Cosmos3-Edge reasoner (Nemotron-2B LM + SigLIP2 tower), SFT on (clip, rubric prompt) → single digit; partial unfreeze (projector + last 8 LM blocks + final norm) fits a single 22 GiB L4. Class-balanced manifest oversampling corrects ordinal imbalance. `[v3]`

Key implementation findings (each cost a debugging cycle; reported for reproducibility): (i) generation-time chat-template thinking mode must be disabled for digit-SFT'd scorers — with it on, every checkpoint rambles CoT and emits no score; (ii) validation loss is the wrong model-selection signal — it bottomed at iter 200 while downstream Pearson kept improving through iter 400; select on task metrics.

3.3 Critic-shaped RL

R = R_task (sparse goal + dense progress) + R_safety (privileged, bounded)

- λ·(critic(clip) − 3)/2, λ bounded. The critic runs **asynchronously**: the env banks episode clips to a file queue; a scorer daemon (owning the GPU) returns scores; bonuses fold into later batches. RL never blocks on the 2B model. Anti-hacking: sparse+privileged terms remain the backbone; periodic audit of top-decile-critic episodes against the auto-labeler.

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4. Experiments

4.1 Can the critic read traversal quality from pixels? (core validation)

Held-out-scene evaluation, ordinal metrics (exact acc, acc±1, Pearson, Spearman, per-class recall).

Model	train clips	val clips	Pearson	acc±1	acc
near-base (20 clips seen)	—	140	0.024	0.557	0.314
critic v1 (best ckpt)	460	140	0.377	0.679	0.457
critic v2 (best ckpt)	1,530	470	**0.482**	**0.755**	0.517
critic v3 (balanced)	2,930	670	0.452	0.743	0.460
critic v4 (clean cam) `[v4]`	~2,900	~590	—	—	—

Scaling is monotone in data and steps through v2. **v3 isolates class balancing** (manifest oversampling to uniform per-score weight): overall correlation holds ≈flat on a harder, larger val while tail-class recall roughly doubles — GT-1 0.22→0.35, GT-5 0.06→0.15, GT-3 0.12→0.24 (Fig. 5). That is the right trade for a reward model: mistaking a dangerous episode for a good one is the costly error, and it is precisely the rare-class confusions that balancing removes. **v4 isolates the camera confound** (§4.4 note): v1–v3 clips blank out the robot for 30–47% of frames in some regimes (chase camera below wall height; wall-fill fraction correlated with score band); v4 regenerates all data with an above-wall view. `[v4]`

4.2 Does critic shaping improve the policy? `[P2 — headline]`

Paired-seed PPO, identical scenes/architecture/steps; arms: (a) sparse+safety baseline, (b) + hand-crafted dense shaping, (c) + critic shaping. Endpoint metrics on held-out scenes: success rate, collision-episode rate, min clearance, person-space violations, time-to-goal, and auto-labeler axis scores (one ruler for policies and critic).

Baseline arm (measured). 300k-step PPO with hand-crafted reward (progress + bounded contact + time). Training-scene success peaked ~0.43 around 112k steps and degraded to ~0.32 by 300k (instability without KL/trust constraints on a small net). On the 40-episode held-out-scene eval:

checkpoint	success	collision-ep rate	mean labeler score
baseline @ peak (112k)	0.15	1.00	1.93

checkpoint	success	collision-ep rate	mean labeler score
baseline @ last (300k)	0.10	0.98	1.83

The train→held-out gap (0.43 → 0.15) and saturated collision rate quantify exactly the headroom the critic arm targets — and note the hand-crafted reward did reach mid-training competence yet produces collisions in essentially every held-out episode, scoring < 2 on the labeler's ruler.

4.3 Real-robot video transfer (measured, preliminary)

We score 42 clips of **real Unitree G1 footage** (GEAR-SONIC release media: in-the-wild navigation, style walks, impaired gait, crawling, kneeling) plus 27 clips from a different simulator, using the best sim-trained critic. No privileged state exists for this footage, so no auto-label can.

Result: **zero parse failures; every clip scored in the 4–5 band** — the correct range, since all release demos are collision-free — with clean walking scoring highest (mean 4.30, the only real regime awarded 5s) and crawl/impaired/posture regimes uniformly at 4.0. The critic transfers to real video without collapsing to noise or misreading clean demos as unsafe. Two honest limits: (i) the released footage contains no negative examples (no real collisions), so this test establishes transfer without collapse, not full discrimination on real video — closing that requires collecting deliberately-flawed real rollouts; (ii) score compression toward 4 mirrors the training distribution's mode.

4.4 Further ablations (planned)

Critic checkpoint quality vs. policy gain; λ sensitivity; band-3 "clean but sloppy" discrimination; real negative-example collection.

4.5 The hand-crafted-reward cautionary tale (motivation, quantified)

First baseline reward: success 0.18 → 0.00 over 150k steps as PPO discovered that standing still dominates walking (per-frame contact penalties × 15 frames/step > max progress term). One line of reward arithmetic, silent collapse. This is the failure mode the critic removes: its judgment is holistic and episode-level, not a sum of per-frame proxies to out-game.

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5. Figures

- **Fig 1 (system)**: three-layer architecture; what deploys vs. what judges. → `paper/figures/fig1_system.svg`
- **Fig 2 (scaling)**: critic Pearson vs. training clips (0.02 → 0.38 → 0.48 → `[v3]`), with acc±1 as secondary panel. → `paper/figures/fig2_scaling.html` (export to PDF/SVG for submission)
- **Fig 3 (qualitative)**: five clips, one per score, with critic score vs. labeler score; egocentric frames strip.
- **Fig 4 (P2 headline)**: paired bars — the three PPO arms on the endpoint metrics.
- **Fig 5 (confusion)**: v2 vs. v3 confusion matrices — the class-balance ablation, visually.

Rendered figures

Fig. 1 — The world model as judge: the critic shapes training-time reward and never deploys.

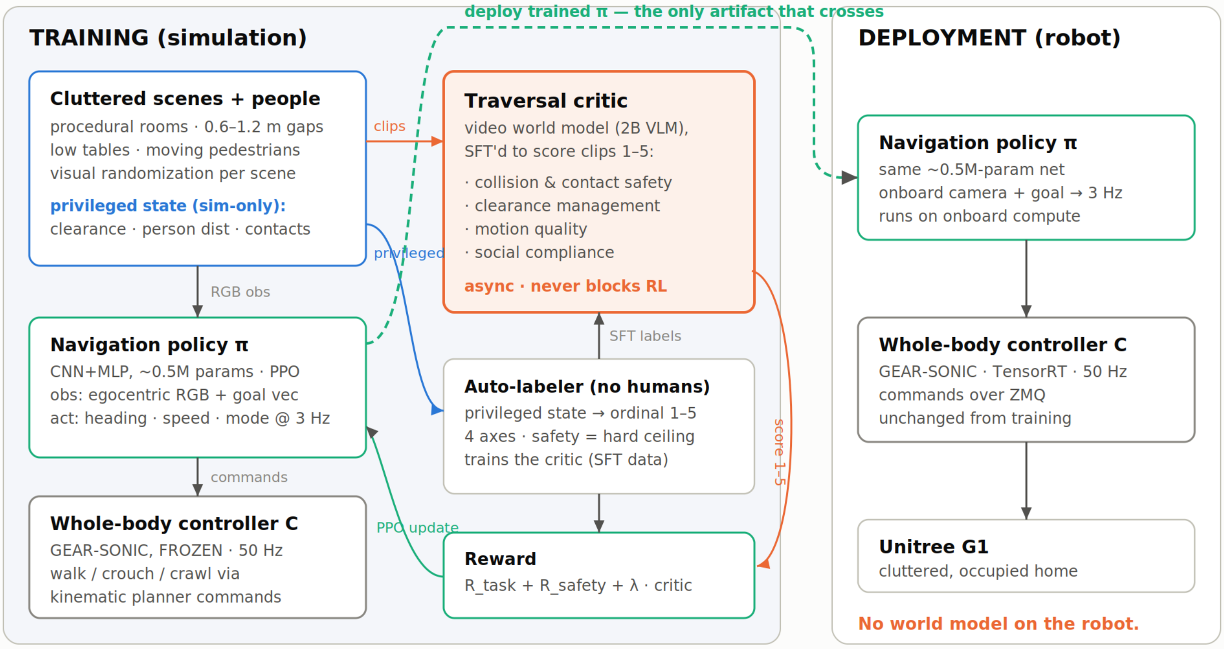
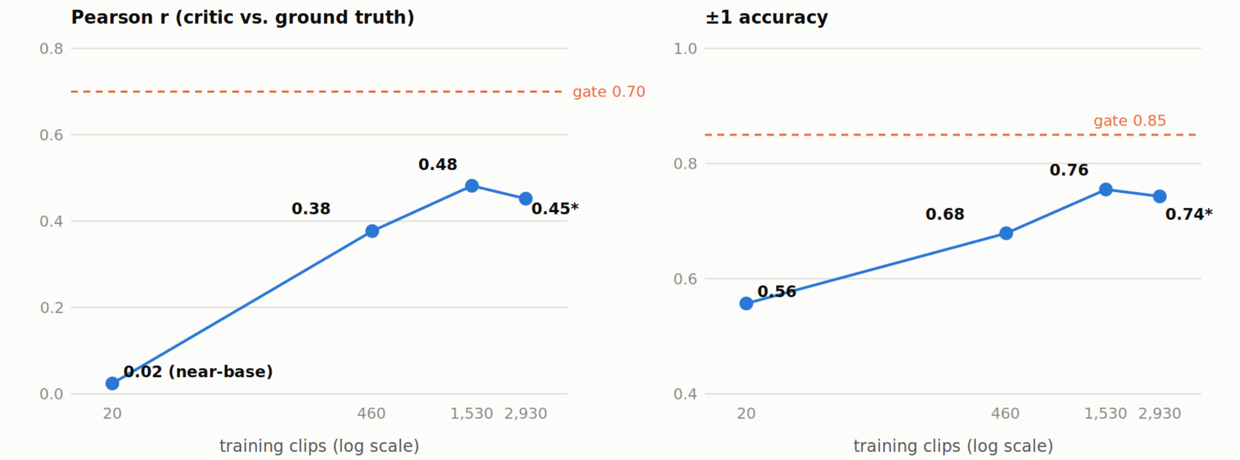


Fig. 2 — The critic scales: traversal-quality prediction on held-out scenes vs. training clips



Scene-level holdout (novel layouts + appearance). Val grows with train (140 → 470 → 670 clips). *v3 is class-balanced: tail recall improves (GT-1 .22→.38, GT-5 .

Fig. 3 — One episode per ground-truth score (frames left to right in time)

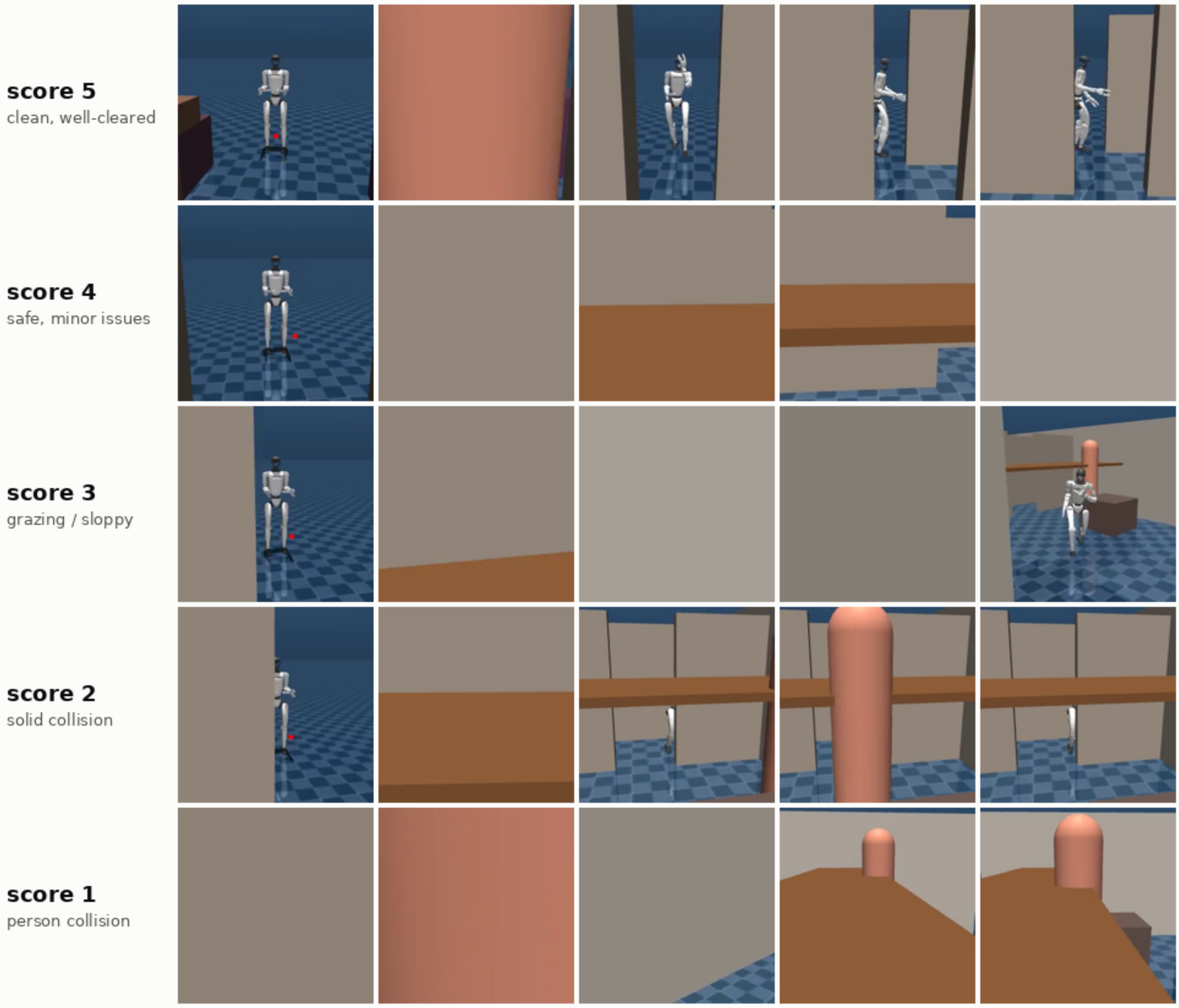


Fig. 5 — Class balancing recovers the rare scores (row-normalized recall)

Fig. 6 — The sim-trained critic transfers to real robot video

42 real G1 clips + 27 other-sim controls · zero parse failures · all scores in the expected band

Critic: v2 iter_400, trained only on procedural MuJoCo clips. Dots jittered; ticks = regime means. No auto-label can exist for this footage.

6. Reproducibility notes

Single shared 22 GiB L4 for everything (critic SFT, eval, scoring); CPU for rollout generation (planner ONNX ~35 ms) and PPO. Full recipe: branch `feat/traversal-critic` — generator, labeler (+18 unit tests), SFT SKU/TOML, eval sweep, scorer daemon, nav env, PPO trainer, paired evaluator.

7. Limitations & staging

Kinematic playback (no physics response) in phase 1–2; SONIC tracker physics-in-the-loop is phase 2-proper (Isaac Lab, pinned 4.5, GRScenes-100 + people layer); real-G1 deployment is phase 4 (ZMQ command path verified in repo docs). Critic gate (Pearson ≥ 0.7) not yet met — scaling curve suggests data, not architecture, is the binding constraint. Auto-labels inherit threshold choices; the human-label override path exists for calibration.